

Purpose

This project works on the development of a serial communication-based calculator using an Arduino Uno microcontroller and a PC console application.

The primary objective is to enable a user to input simple arithmetic expressions through the PC application, which is then transmitted to the Arduino over a USB serial connection. The project in its final phase added some additional features like being able to type in the communication port instead of hard coding it and additional operator like the power operator, root, and being able to get remainder of two numbers.

Mode of Operation

This system enables bidirectional communication between the Arduino and the PC via a USB serial interface. The user inputs arithmetic expressions through the console application, which transmits the expressions to the Arduino for processing. The Arduino parses the input and performs the requested arithmetic operation and returns the computed result or an appropriate error message back to the console app. After a set of arithmetic operations are performed by the user and are deemed complete, the history can then be logged to a .txt file (results.txt) in the project folder by using the phrase "exit". This .txt file provides information on the date, time, the operation and the feedback, either a result or an error. This logs all the arithmetic operation activity, successful or not, performed by the user in as much as the said phrase is used.

Supported Operators

The following are the supported operators in the app as of the finalization phase:

1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (^)
6. Root ($n\sqrt{}$)
7. Remainders (&)

Technologies Used

1. This system uses C++ for the programming language of both the PC console app and the Arduino code.
2. Uses the UART communication protocol between the microcontroller and the PC
3. Uses the SerialPort library cited in documentation of the app.
4. Can be compiled into an executable (.exe) format via visual studio or directly from the command line.

Important Decisions and Solutions

The design enables separation of the user interface and the processing unit or computational logic, input validation by using defining a consistent input format (<number1> <operator> <number2>) and error handling. When errors are detected due to the user input, a pre-error message is displayed, and this message and the reason for the message can be checked in the user manual.

The Arduino Uno has limited memory and processing speed, making it difficult to handle complex string operations or long inputs. This made it difficult to use the `sscanf` function to parse the input string to numbers and operator because it keeps reading 0. Instead, I had to manually parse the string for floats.

In addition, I implemented a change in the Console app that allows the user to manually input the COM port at the beginning of runtime instead of having to hardcode it into the Console app code. And then also fixed the issue of synchronization between the input, processing and output.

Additionally, there is modularity. By separating the `SerialPort`, `expression parsing` and `calculatorapp`, readability is improved and can be expanded upon in the future if desired.

In summary, this project through a PC console application with an Arduino Uno was made to create a simple serial-based arithmetic calculator. The system separates user interaction from processing logic with communication between the interface and microcontrollers. Key challenges were input validation, serial synchronization between the input, processing and then output, error handling.

GitHub Link to Project: <https://github.com/Mexperse/Serial-Calculator.git>